



# Architecture Document Structure

हम Phase 3 को भी Professional तरीके से बनाएँगे।

## Chapter 1 – System Overview

### 1.1 Purpose

The purpose of this document is to define the complete software architecture of the **Gold Trading Automation Platform (GTAP)**. It explains how each module communicates with other modules and how data flows throughout the system.

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### 1.2 Architecture Goals

The architecture is designed to achieve:

- High Reliability
  - Low Latency
  - Easy Maintenance
  - High Security
  - Scalability
  - Modular Development
  - Real-time Trading
  - Fault Tolerance
- 

### 1.3 Technology Stack

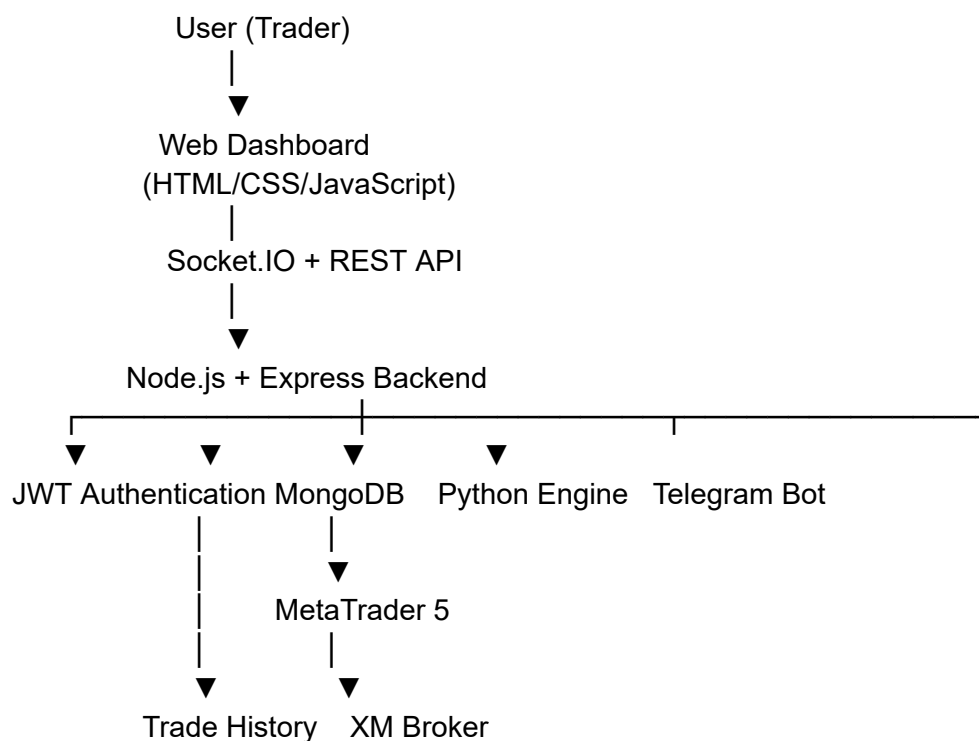
| Layer            | Technology            |
|------------------|-----------------------|
| Frontend         | HTML, CSS, JavaScript |
| Backend          | Node.js + Express.js  |
| Database         | MongoDB               |
| Trading Engine   | Python                |
| Trading Platform | MetaTrader 5          |
| Broker           | XM                    |
| Authentication   | JWT                   |

|                   |                          |
|-------------------|--------------------------|
| Notification      | Telegram Bot API         |
| Deployment        | Ubuntu VPS + PM2 + Nginx |
| Real-Time Updates | Socket.IO                |

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## Chapter 2 – High-Level Architecture

पूरा सिस्टम इस तरह काम करेगा:



## Explanation

### Frontend

जिम्मेदारी:

- Login
- Dashboard
- Reports
- Settings
- Live Trade Monitoring
- Bot Start/Stop

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## ● Backend

यह पूरे सिस्टम का **Brain** होगा।

Responsibilities:

- Login
- Authentication
- API
- Dashboard
- Database
- Python Communication
- Telegram
- Logging

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## ● Python Trading Engine

यह सबसे Important Module होगा।

Responsibilities:

- MT5 Connection
- EMA Calculation
- RSI Calculation
- Fractal Detection
- Buy/Sell Decision
- Order Execution
- SL
- TP
- Trailing SL

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## ● MongoDB

Store करेगा:

- Users
  - Trades
  - Reports
  - Logs
  - Bot Settings
-

## ● MT5

- Market Data
  - Execute Trade
  - Account Information
- 

## ● Telegram

- Buy Alert
  - Sell Alert
  - Error Alert
  - Daily Report
- 



# Chapter 3 – Architecture Principles

हम Project को इन Principles पर बनाएँगे:

## Modular Design

हर Module अलग होगा।

Example:

- Dashboard
  - Trading
  - Database
  - API
- 

## Loose Coupling

Python को Frontend के बारे में पता नहीं होगा।

Frontend को MT5 के बारे में पता नहीं होगा।

हर Module केवल Backend से बात करेगा।

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## High Cohesion

हर Module अपना एक Specific काम करेगा।

Example:

Trading Engine सिर्फ Trading करेगा।

Database सिर्फ Data Store करेगा।

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## Security First

- JWT
  - Password Hashing
  - Secure APIs
  - Environment Variables
- 

## Scalability

आज:

- 1 User

कल:

- 100 Users

Architecture बदलने की जरूरत नहीं होगी।

# Chapter 4 – Low-Level Architecture

## 4.1 Frontend Layer

### Technology

- HTML5
- CSS3
- JavaScript
- Socket.IO Client
- Axios (API Calls)

### Responsibilities

- User Login
- Dashboard
- Live Profit/Loss
- Active Trades

- Trade History
- Reports
- Settings
- Bot Start/Stop
- Real-time Updates

## Communication

Frontend



REST API + Socket.IO



Node.js Backend

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## 4.2 Backend Layer

### Technology

- Node.js
- Express.js
- JWT
- Socket.IO
- Axios
- Mongoose

### Responsibilities

- Authentication
  - API Routing
  - Dashboard Data
  - MongoDB Operations
  - Python Communication
  - Telegram Notifications
  - Error Handling
  - Logging
- 

## 4.3 Trading Engine

### Technology

- Python
- MetaTrader5 Library

## Responsibilities

- MT5 Login
  - Candle Reading
  - EMA Calculation
  - RSI Calculation
  - Fractal Detection
  - Buy Logic
  - Sell Logic
  - Stop Loss
  - Take Profit
  - Trailing Stop
  - Position Monitoring
- 

## 4.4 Database Layer

### MongoDB Collections

- Users
  - Trades
  - Reports
  - Logs
  - Settings
  - Sessions
- 

## 4.5 Notification Layer

Technology:

- Telegram Bot API

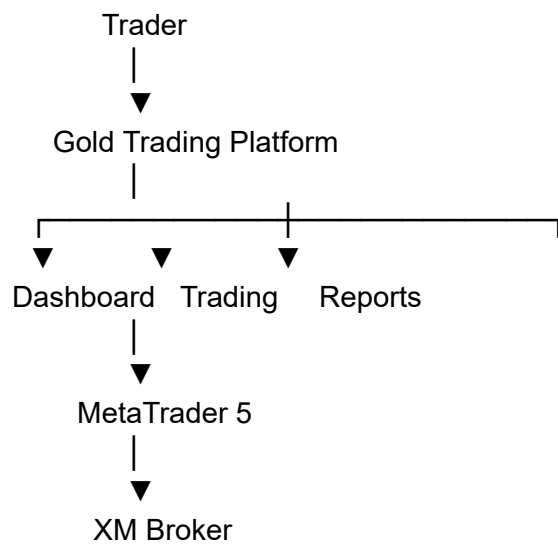
Responsibilities:

- Buy Alert
  - Sell Alert
  - Error Alert
  - Daily Report
  - Bot Status
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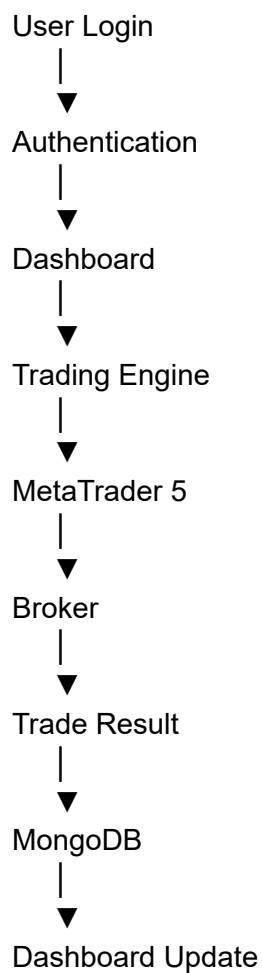


## Chapter 5 – Data Flow Diagram (DFD)

## Level 0 DFD



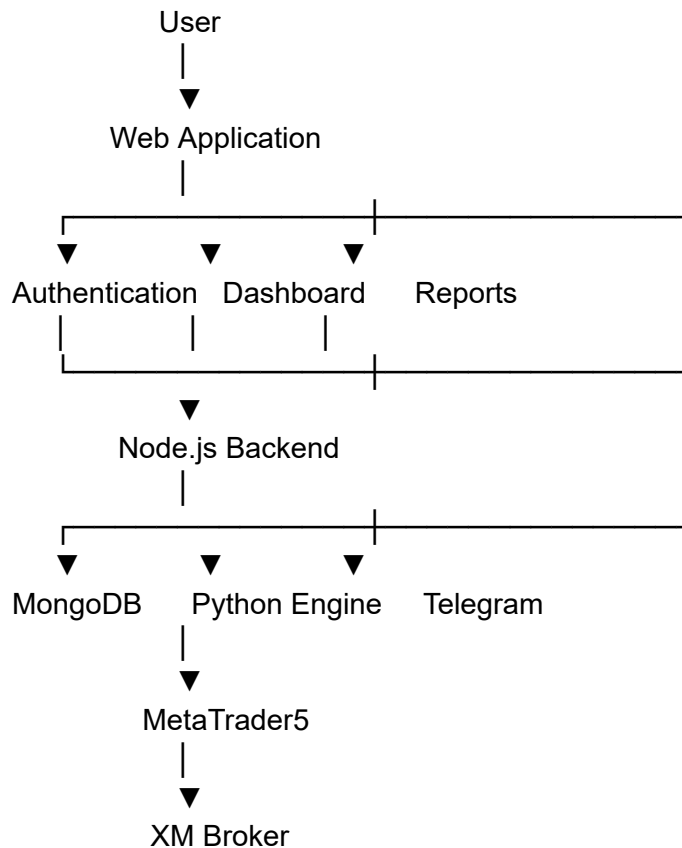
## Level 1 Data Flow





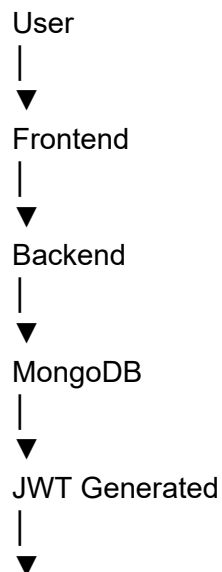


## Chapter 6 – Component Diagram



## Chapter 7 – Sequence Diagram

### Login Process



Frontend

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Dashboard

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## Buy Trade Process

Python Engine



Reads Candle



EMA Calculation



RSI Calculation



Fractal Detection



Strategy Check



Buy Signal



MT5 Order



MongoDB Save



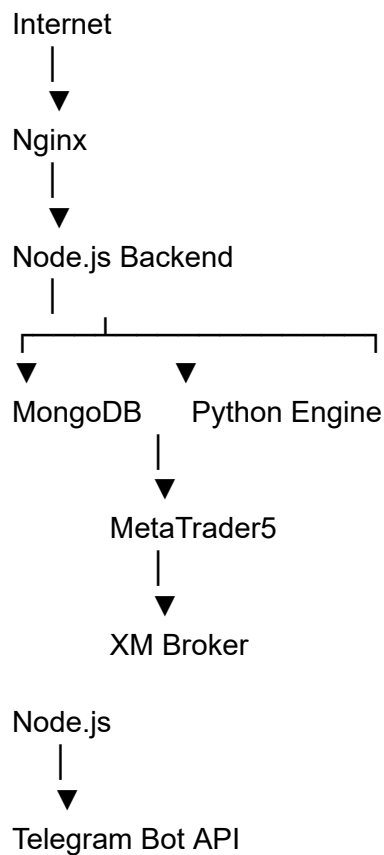
Telegram Alert





## Chapter 8 – Deployment Architecture

### Production Environment



### Deployment Components

| Component        | Technology    |
|------------------|---------------|
| Operating System | Ubuntu Server |
| Web Server       | Nginx         |
| Backend          | Node.js       |
| Trading Engine   | Python        |
| Database         | MongoDB       |
| Process Manager  | PM2           |

|                  |              |
|------------------|--------------|
| Trading Platform | MetaTrader 5 |
|------------------|--------------|

|               |              |
|---------------|--------------|
| Notifications | Telegram Bot |
|---------------|--------------|